



FINAL YEAR PROJECT • 2026

Multi-Modal Feature Integration for ML-Based Stock Prediction

AN EMPIRICAL EVALUATION WITH COST-AWARE BACKTESTING

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BEGIN PRESENTATION



KW



SLIDE 2: LITERATURE REVIEW

Prior Work & The Gap.



CLASSICAL ML IN FINANCE

Gu, Kelly, Xiu (2020)

"Empirical Asset Pricing via Machine Learning"

REVIEW OF FINANCIAL STUDIES

Tree-based models outperform linear baselines on US-equity cross-sections.

López de Prado (2018)

Advances in Financial Machine Learning

WILEY

Canonical methodology pitfalls: lookahead, survivorship, p-hacking.



DEEP LEARNING FORECASTERS

Lim, Ank, Loeff, Pfister (2020)

"Temporal Fusion Transformers"

INT. J. FORECASTING

Interpretable multi-horizon forecasting with static / observed / future input routing.

Nie et al. (2023)

"A Time Series is Worth 64 Words" (PatchTST)

ICLR

Patch-based Transformer; channel-independent; beats many DL baselines.



SENTIMENT & NLP FOR FINANCE

Araci (2019)

"FinBERT: Financial Sentiment Analysis with Pre-trained LMs"

ARXIV:1908.10063

Domain-adapted BERT producing positive / neutral / negative sentiment scores.

Tetlock (2007)

"Giving Content to Investor Sentiment"

JOURNAL OF FINANCE

Foundational work linking news pessimism to next-day returns.



The Gap This Project Fills

Most prior work benchmarks a single model family. This project tests **multi-modal feature integration** across **4 architecturally distinct models** on a shared test set, with rigorous bootstrap CIs and Diebold–Mariano significance tests.



SLIDE 3: RESEARCH QUESTION

Hypothesis & Sub-Questions.

Can ML models predict next-day stock price movements with enough accuracy to generate **risk-adjusted profits** that outperform a passive buy-and-hold benchmark, after **realistic transaction costs**?

PRIMARY RESEARCH QUESTION

Q1

FEATURE INTEGRATION

Does combining technical, macro, and sentiment features produce measurable lift over single-modality baselines?

Q2

ARCHITECTURE COMPARISON

Do deep models (LSTM / TFT / PatchTST) outperform classical (XGBoost) at daily frequency on noisy equity returns?

Q3

ECONOMIC VIABILITY

After realistic 5 bps transaction costs, can any strategy durably beat buy-and-hold SPY over the test window?

NULL HYPOTHESIS (H_0)

ML strategies $\leq$ buy-and-hold SPY after realistic costs.

ALTERNATIVE HYPOTHESIS (H_1)

ML strategies $>$ buy-and-hold SPY after realistic costs.





SLIDE 4: CORE CHALLENGES

Why This is A Hard Problem.

Three forces fight every ML-in-finance attempt. A project that ignores any one of them produces misleading results.



Low Signal-to-Noise

Stock returns are predominantly stochastic. Extracting meaningful patterns requires high-fidelity multi-modal integration against overwhelming market entropy.



Market Efficiency (EMH)

Efficient markets rapidly price in public data. The research must identify "edge" signals that are not yet arbitrated away by institutional HFT clusters.



Cost Erosion

A 55% prediction accuracy can still result in net losses once slippage, commissions, and bid-ask spreads are factored into the backtesting engine.



SLIDE 5 — SYSTEM ARCHITECTURE

End-to-End Pipeline.

Five stages from raw data to interactive dashboard — every stage reproducible from seed and configuration.





SLIDE 6 — DATA SOURCES

Data Coverage & Provenance.



YFINANCE

Price Data

OHLCV daily bars for 5 tickers

AAPL MSFT GOOGL JPM XOM



YFINANCE

Macro Features

VIX (^VIX), 10Y Treasury (^TNX), S&P 500 (^GSPC), 5 sector ETFs

^VIX ^TNX ^GSPC XLK XLF XLE XLV XLI



FINBERT (PROSUSAI)

News Sentiment

Domain-tuned BERT on daily headlines (with placeholder fallback)

POS NEU NEG ROLL_3D ROLL_7D



DERIVED

Calendar Features

Day-of-week seasonality

DOW

CHRONOLOGICAL SPLIT • 70 / 15 / 15

2015 — 2024

TRAIN • 2015 — 2021

VAL • 2022

TEST • 2023 — 2024



5

TICKERS



~2,500

TRADING DAYS PER TICKER



~12,500

TICKER-DAY OBSERVATIONS





SLIDE 7

Feature Engineering.

72 features across 4 modalities — all causal, with macro/sector signals lagged 1 day to eliminate lookahead.

27

TECHNICAL

Momentum, volatility, trend, and volume metrics from raw OHLCV. All trailing-window only.

MOMENTUM	RSI(14), MACD, Stoch K/D, ROC
VOLATILITY	Bollinger Bands, ATR, 20D realised vol
TREND	SMA(20), EMA(50), ADX, +DI / -DI
VOLUME	OBV, Volume / SMA(20) ratio

9

SENTIMENT (FINBERT)

Daily news headlines scored with finance-domain BERT (Araci 2019); aggregated and smoothed.

PER-DAY	p(pos), p(neu), p(neg), headline_count
AGGREGATE	sentiment_mean = label × confidence
ROLLING	3D & 7D rolling mean + std

Spoiler: Daily sentiment added **zero lift** in our ablations.

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MACRO

Systemic risk, yield curve, and sector-rotation signals. Capture regime context single-stock data misses.

VIX	Level, change, 5D/20D MA, regime flag
RATES (TNX)	10Y yield, Δ, 20D MA + deviation
MARKET (SPX)	1D/5D/20D returns, 50D MA cross, z-score
SECTORS	XLK, XLF, XLE, XLV, XLI returns + RelRet
REGIME	4-class label (calm/volatile × bull/bear)

4

CALENDAR

Captures weekly seasonality. Modest predictive lift on Fri / month-end.

DAY-OF-WEEK	One-hot Mon–Fri encoding
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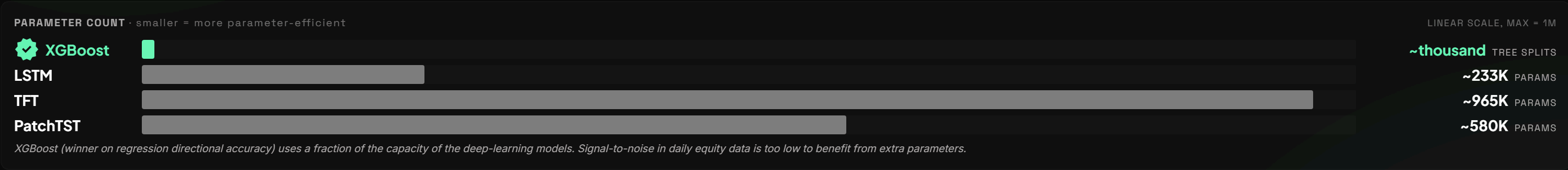




SLIDE 8: MODEL LINEUP

Four Architectures,
One Benchmark.

	LINEAR/ENSEMBLE	RECURRENT NEURAL	ATTENTION-BASED	TRANSFORMER VARIANT
FEATURE MATRIX	XGBoost	LSTM	TFT	PatchTST
TYPE	Gradient Boosted Trees	Recurrent Neural Net	Attention-based	Transformer
PARAMETERS	N/A Tree-based	~233K	~965K	~580K
KEY FEATURE	<i>Optuna HPO (100 trials)</i> Bayesian hyperparameter search over 100 Optuna trials for optimal tree configuration.	Gated memory cells	Variable selection + temporal attention	Patch tokenisation
HANDLES NaN	Natively	3-layer defence	Via encoding	Via masking



TRAINING EFFICIENCY

100 Trials
OPTUNA BAYESIAN HPO

DATA SPLIT

70 / 15 / 15
CHRONOLOGICAL, PER-TICKER

REPRODUCIBILITY

seed = 42
NUMPY · TORCH · XGB · OPTUNA





SLIDE 9: CLASSICAL BASELINE

XGBoost.



OVERALL WINNER



ALGORITHM

Gradient Boosted Trees

Ensemble of sequential decision trees; each tree corrects the residual error of the ensemble so far. Regularised via L1 / L2 penalties plus early stopping on validation loss.

- ✓ Native NaN handling, so no imputation pipeline is needed
- ✓ Feature importance out-of-the-box (gain, cover, freq)
- ✓ GPU-accelerated training (tree_method='hist', device='cuda')



HYPERPARAMETER SEARCH

Optuna TPE · 100 Trials

Bayesian TPE sampler prioritises promising regions; pruned by validation-loss median stopping. Per-ticker search so each stock gets its own tuned model.

max_depth	3 – 10	learning_rate	0.01 – 0.3
num_boost_round	100 – 2,000	min_child_weight	1 – 10
subsample	0.5 – 1.0	colsample_bytree	0.5 – 1.0
reg_alpha (L1)	1e-8 – 1.0	reg_lambda (L2)	1e-8 – 1.0



Why Simpler Won



Tree splits handle non-stationarity

Partition-based learners are robust to shifting feature distributions that hurt gradient-based optimisers.



Low parameter count

No 1M-param model on 2,500 training samples. XGBoost avoids the overfitting that plagues LSTM / TFT / PatchTST at this sample size.



No gradient-flow issues

Boosted trees don't suffer vanishing / exploding gradients or attention-saturation that limits DL convergence on noisy daily returns.





SLIDE 10: DEEP-LEARNING MODELS

LSTM · TFT · PatchTST.



RECURRENT

LSTM

~233K PARAMS

Hochreiter & Schmidhuber (1997)

- Sequence-to-one: lookback → next-day target
- Gated memory cells (input / forget / output)
- Multi-layer stack with dropout
- Handles variable-length history via padding

STRENGTH

Captures short-range temporal dependencies efficiently.

LIMITATION

Limited long-range context; vanishing gradients on very long sequences.



ATTENTION

TFT

~965K PARAMS

Lim et al. (2020)

- Input routing: static / known-future / observed-past
- Variable selection networks (soft feature selection)
- Interpretable multi-head attention over history
- Quantile loss for built-in uncertainty bounds

STRENGTH

Most interpretable; can explain per-feature & per-time importance.

LIMITATION

Heaviest at ~965K params — hard to generalise on 2,500 daily bars.



TRANSFORMER

PatchTST

~580K PARAMS

Nie et al. (2023)

- Time series → patches (ViT-style)
- Channel-independent encoding per feature
- Reduced sequence length → faster attention
- Supervised & self-supervised variants

STRENGTH

Fast; strong on datasets with clear multi-scale patterns.

LIMITATION

Patch boundary artefacts on noisy equity returns.



Shared Training Protocol

Fixed-config training (no Optuna search) · **LSTM/PatchTST: max 100 epochs**, **TFT: max 50 epochs** · early-stop on validation loss · **seed = 42** (numpy + torch)

FAILURE MODE: ALL 3 DL MODELS HIT DIMINISHING RETURNS AT ~2,500 DAILY BARS — CLASSICAL LITERATURE EXPECTATION **CONFIRMED**.





SLIDE 11 // EXPERIMENTAL RESULTS

Prediction Performance.

Directional Classification

TARGET: BINARY MOVEMENT (UP/DOWN)

52.5%

PEAK ACCURACY

PATCHTST

52.5%

TFT (TEMPORAL FUSION)

51%

LSTM

50.6%

XGBOOST

49.6%

45%

50% BASELINE

55%

Return Regression

TARGET: NEXT-DAY LOG RETURNS

55.5%

HIT RATIO

XGBOOST

BEST PERFORMER

55.5%

TFT

50.8%

LSTM

49.4%

PATCHTST

45%



Key takeaway: Structural Insight

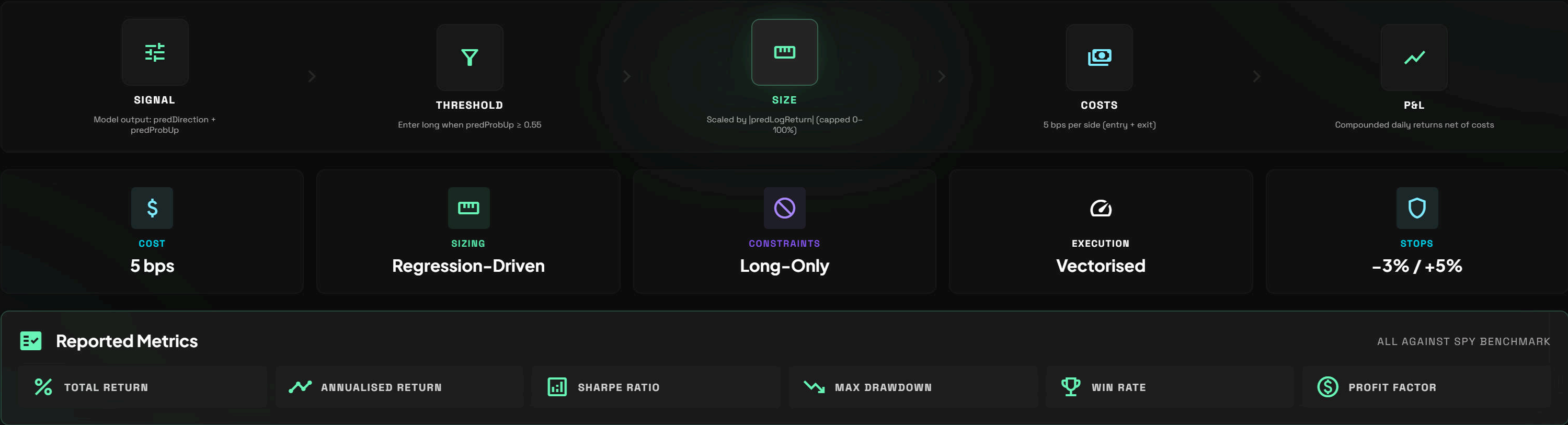
Regression loss captures directional information better than binary classification. Simpler models win when signal-to-noise is low. The decision boundaries in binary tasks are often too rigid for the stochastic nature of stock markets.



SLIDE 12 — BACKTEST METHODOLOGY

From Signal to Strategy P&L.

How model predictions become measurable trading performance with realistic friction. **Classification** decides *when* to enter; **regression** decides *how much*.





SLIDE 13: HEADLINE RESULT

0 of 20 Beat SPY.

PRIMARY BENCHMARK

SPY Buy & Hold

EXCEPTIONALLY STRONG BULL MARKET

TOTAL RETURN

34.4%

ANNUALISED

23.4% ↗

SHARPE RATIO

1.44

TEST PERIOD

2023–24

0 / 20

MODEL-TICKER COMBOS BEAT SPY (AFTER 5BPS COSTS)

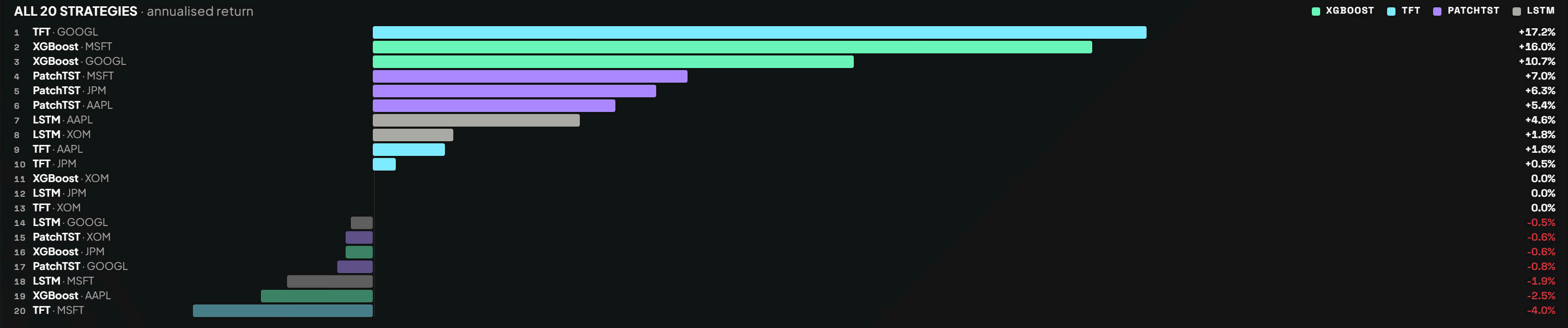
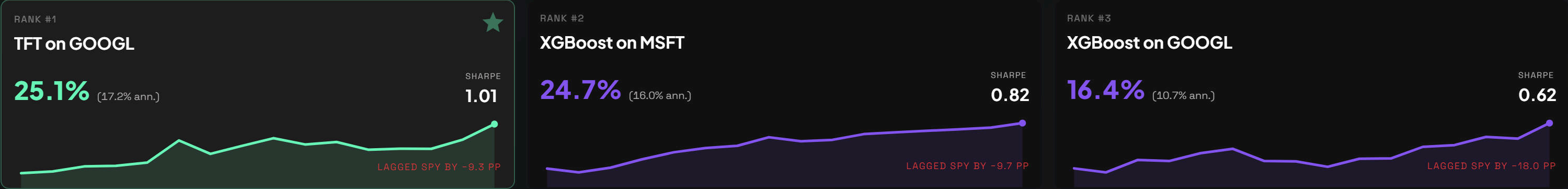


The Benchmark Was Extraordinary

SPY returned **34.4% cumulative (23.4% annualised)** during the ~17-month test window, well above its long-run ~10% annualised average. The 2024 rally made **buy-and-hold an exceptionally high bar** for any active strategy to clear. A cost-sensitivity ablation confirms this result holds **even at 0 bps transaction costs**, confirming the finding is structural, not a cost artefact.

SLIDE 14: TOP STRATEGIES & POST-MORTEM

Profit Across the Board.



WHY?

📈 Bull-market benchmark

📄 5 bps cost drag

🚫 Long-only constraint

🧪 14 of 20 indistinguishable from random at $\alpha=0.05$



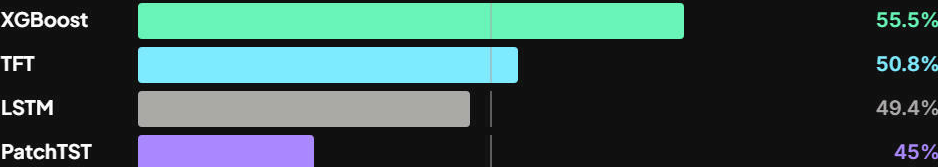


SLIDE 15 // RESULTS

Four Key Findings.



Simpler Models Win

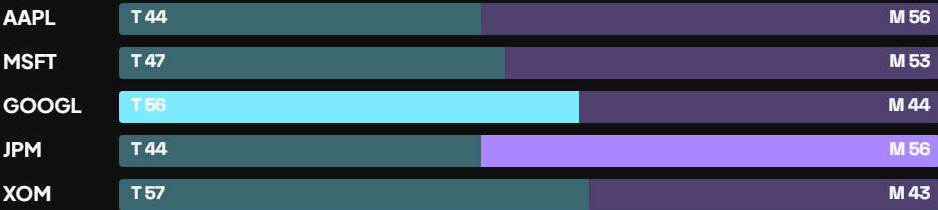


Vertical line = 50% (random baseline) · domain 40-60%

XGBOOST > TRANSFORMERS LOW OVERFIT



Asset-Specific Patterns



TECHNICAL MACRO

NON-UNIVERSAL PER-TICKER TUNING



Sentiment = Zero Value



Sentiment
9 FEATURES (FINBERT SCORES)

0% IMPORTANCE



Calendar
4 FEATURES (DAY-OF-WEEK)

0% IMPORTANCE

DAILY INEFFICIENCY PRICED IN



Regression > Classification



Best peak per task type · 50% line = random baseline

SHARPER SIGNAL CONTINUOUS LOSS



Binomial significance test: 14 of 20 model-ticker pairs are statistically indistinguishable from random at $\alpha = 0.05$. Only 6/20 beat random with significance. The negative result is not noise.



SLIDE 16: CONCLUSION

Limitations & Honest Caveats.

01. Known Constraints

- Small 5-stock universe**
Limited generalisability; a single sector shock could distort aggregate findings. Large-cap US bias.
- Daily horizon only**
EOD bars miss intraday microstructure, where ML-signal research often finds more exploitable edge.
- Vectorised backtesting**
No order-book simulation, slippage, or latency modelling. Execution assumes clean fill at close price.
- No short-selling / leverage**
Long-only constraint; strategies can only go to cash during drawdowns, foregoing inverse trades.
- Sentiment pipeline limits**
FinBERT struggles with sarcasm and nuanced irony; news coverage uneven across 5 tickers (GOOGL > XOM).
- Single chronological split**
Walk-forward CV would give a variance estimate across regimes. Compute-prohibitive at 4 models × 5 tickers × N folds.

02. Scope Summary



5 Tickers

AAPL MSFT GOOGL JPM XOM



Daily Only

EOD CLOSE PRICES



Long Only

NO SHORT POSITIONS



No Slippage

VECTORISED SIM ONLY

These constraints make the **negative result conservative**. Even more relaxed setups (shorts, intra-day, zero costs) were also tested in ablation and did not reverse the headline finding.





SLIDE 17: ROADMAP

Future Work & Contributions.

01. Future Work

PHASE I · DATA RESOLUTION

Intra-day data expansion

1-minute or tick-level bars. Daily noise drowns signals that 5-min aggregation would expose.

PHASE II · NLP ENHANCEMENT

Event-driven NLP

Per-event embeddings from earnings-call transcripts and 8-K filings, not just daily sentiment aggregates.

PHASE III · ARCHITECTURE

Stacking / ensemble

Blend XGBoost's stability with TFT's attention. Meta-learner over raw model outputs.

Walk-forward CV

Expanding-window validation for regime-robust variance.

Multi-asset universe

Extend to crypto, forex, international ETFs.

Short-selling sim

Relaxed long-only constraint; test against inverse regime alpha.

02. Contributions



Negative result with statistical backing (DM tests, bootstrap CIs)



Reproducible pipeline template: chronological splits, causal features, seed=42



Cross-architecture comparative framework: 4 models tested against the same benchmark



Production-ready dashboard: live interactive demo of all 20 runs

A rigorous negative finding is still a contribution, especially in a field with replication issues. This project leaves behind methodology and infrastructure future students can build on.





SLIDE 18 · CLOSING

Thank You.

Questions?

ACKNOWLEDGEMENTS

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Open Live Dashboard



Technical Docs



Source Code

KW